

Mathematical Modeling of Drone Applications for Wildfire Detection and Research

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Problem Statement

The lack of rapid, efficient wildfire response and measurement of post-wildfire's environmental impact, which could be resolved through the implementation of drone technology

Objective

Utilizing the City of Irvine as a model, we aim to identify the optimal drone launch location and maximize coverage while ensuring efficient disaster response operations. Additionally, we aim to find practical methods to apply our drone network to collect academic and field-relevant data surrounding large-scale wildfires, whether pre- or post-event. Our object is to develop mathematical models and algorithms that optimize the deployment and operation of drones in disaster scenarios, specifically wildfires. The real-world constraints such as air resistance, battery life, and winds will be considered and calculated to make it possible to apply in real-world situations.

Background

Drones have been used previously in disaster relief efforts such as during the 2017 Hurricane Harvey, or the 2020 California Wildfires. Drones have been used to aid rescue efforts by assessing damage and delivering small items for immediate relief. Scientific papers such as *Drone Applications for Emergency and Urgent Care: A Systematic Review* state that, "One of the main benefits of drones is to avoid endangering rescuers in shootings, fires, radiation, the presence of infectious agents, explosives, smoke, or gases." Drones are small, fast, and move in all 3D directions, allowing it to reach areas rescuers would normally not be able to reach.

Drones can be used to collect post-wildfire data, such as information on biological changes in areas that are inaccessible to researchers or located in disaster zones. Although drones have been used in previous disaster relief efforts, there have been no systematic approaches to the efficient deployment of drones to minimize damage to nature and property. We seek to remedy that and establish an efficient grid of drone technology to respond as quickly as possible and reduce damage. The recent Palisades and Eaton Fire is proof of this; the lack of rapid response contributed to the already disastrous effects of the fires. By utilizing new technologies like AI modeling, LiDAR, and UAS, drones can be heightened to new levels of efficiency and collect important information to fight fires as well as other natural disasters.

Theoretical Framework & Notation

Drag Force (Air Resistance): The force opposing a drone's motion due to air, given by

$$F_{\text{drag}} = \frac{1}{2} \rho C A v^2$$

Since $F = m \cdot a$, we can substitute F_{drag} .

$$m \cdot a_{\text{drag}} = \frac{1}{2} \rho C A v^2$$

where

ρ = air density (1.293 kg/m^3)

C = drag coefficient (0.5117)

A = cross-sectional area (0.5427 m^2)

v = velocity (m/s)

M = mass of the drone (3.6 kg)

This drag force will be plugged into an equation to generate a final differential equation, which is then solved and utilized throughout the results.

The most efficient distance: The shortest distance between the most hazardous regions is calculated by:

Geometric Median

$$x^* = \arg \min_x \sum ||x - x_i||$$

where $||x - x_i||$ is the Euclidean distance between the candidate location x and the

hazard zone x_i

In this research, the flight duration for all drones is kept constant. Given uniform wind speed and direction, each drone will experience identical displacement during flight. Therefore, the required procedure involves using the battery drainage time to determine the displacement caused by the wind using the equation $d = vt$. Since this displacement will be the same for all drones, the launch site should be shifted uniformly in the opposite direction by the same magnitude to compensate accordingly.

Battery Efficiency Model

Battery performance is affected by temperature and power consumption. The model used is:

$$T_b = \frac{C \cdot \eta}{P_{total}}$$

T_b = Battery lifetime

C = Battery capacity

η = Battery efficiency factor (varies with temperature),

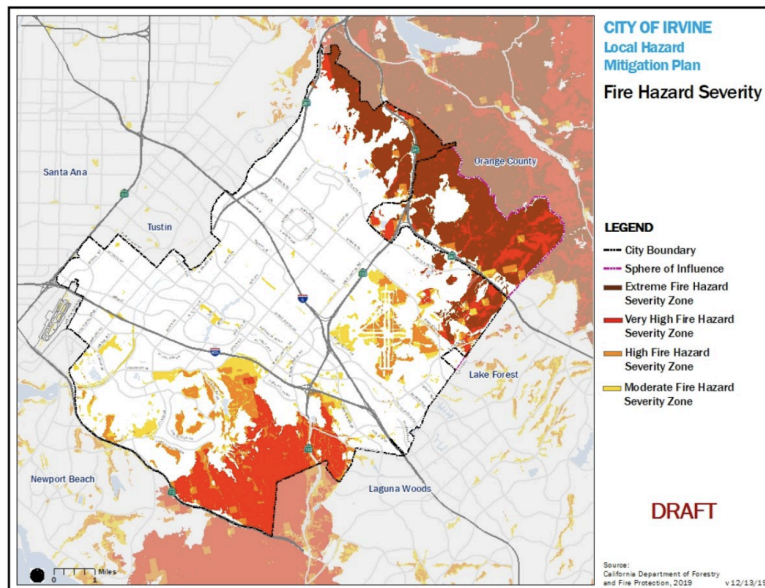
P_{total} = Total power consumption.

Accounting for altitude changes, we estimate battery performance under realistic operating conditions.

Methodology

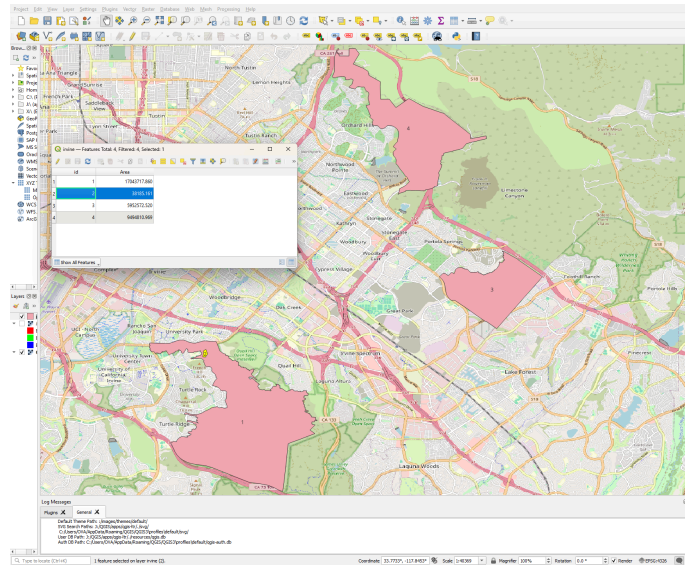
1) Data Collection

- Utilize Geographic Information System (GIS) tools to map high-risk wildfire zones in Irvine CA



→ Find the exact area of Irvine and high risk fire zones in km^2

- Extract coordinate data for fire hazard areas from the Orange County Fire Authority Fire Hazard Severity Zones map



- Record wind speed and direction (8mph, westward, assumed from spring season.)
- Obtain drone specifications such as flight, battery life, and wind resistance (using DJI Matrice 300 RTK drone data)

```

1  import numpy as np
2  from scipy.spatial.distance import cdist
3
4  points = np.array([
5      [33.62057, -117.79181],
6      [33.654105, -117.807299],
7      [33.67728, -117.69881],
8      [33.72948, -117.72014]
9  ])
10
11
12  initial_guess = np.mean(points, axis=0)
13
14
15  def geometric_median(points, initial_guess, max_iter=100, tol=1e-6):
16      guess = initial_guess
17      for _ in range(max_iter):
18
19          distances = cdist([guess], points)[0]
20
21
22          if np.any(distances == 0):
23              return points[np.argmin(distances)]
24
25
26          weights = 1 / distances
27
28
29          new_guess = np.sum(weights[:, None] * points, axis=0) / np.sum(weights)
30
31
32          if np.linalg.norm(new_guess - guess) < tol:
33              return new_guess
34
35          guess = new_guess
36      return guess
37
38  geometric_median_point = geometric_median(points, initial_guess)
39
40  print(f"The geometric median is at: {geometric_median_point}")
41

```

- The geometric median is at: [33.66344288 -117.76359423]
- The program used to find the geometric median
-

2) Mathematical Modeling and optimization

- Develop equations considering air resistance, wind effects, and battery limitations.

Find the displacement of the drone given time

Let a be the acceleration of the drone; let's say $a = 5 \text{ m/s}^2$

We need to solve for the drag coefficient to use the air resistance formula with a given constraint

$$\rho = 1.293 \text{ kg/m}^3$$

$$A = 0.5427 \text{ m}^2$$

$$A_{\text{new}} = 0.1 \text{ m}^2$$

$$M = 3.6 \text{ kg}$$

$$ma = 1/2 \rho C_D A v^2$$

$$C_D = \frac{2ma}{\rho v^2 A_{\text{new}}} = \frac{2(3.6)(5)}{(1.293)23^2(0.1)} = 0.5117$$

Then, we calculate the acceleration

$$a = \frac{\rho C_D A v^2}{2m} = \frac{(1.293)(0.5117)(0.1)v^2}{2(3.6)} = 0.00919$$

To take the air resistance into account, we must add the acceleration of the air resistance.

$$\frac{dv}{dt} = -0.00919v^2 + 5 ; \text{ or simply } \frac{dv}{dt} = 5 - rv^2 \text{ where } r = 0.00919$$

$$\frac{dv}{5 - rv^2} = dt$$

$$t = \frac{1}{5} \sqrt{\frac{5}{r}} \operatorname{arctanh}\left(\sqrt{\frac{r}{5}} v\right)$$

$$v = \sqrt{\frac{5}{r}} \tanh\left(5t * \sqrt{\frac{r}{5}}\right)$$

However, the drone will reach its maximum speed of 23 m/s

So, the time it takes to reach the velocity of 23 m/s is

$$23 = \sqrt{\frac{5}{r}} \tanh\left(5t * \sqrt{\frac{r}{5}}\right)$$

$$t = 11.5663$$

During the first 11.5663 seconds of the flight, the drone will fly

$$\int_0^{11.5663} \left(\sqrt{\frac{5}{r}} \tanh(5t * \sqrt{\frac{r}{5}}) \right) dt = 195.1m$$

After the acceleration process, the drone will cruise at the maximum speed of 23 m/s

So, the distance traveled by the drone in a given time is

$$d = 195.1 + 23(t - 11.5663) \text{ for } t \geq 11.5663$$

We can use the equation we found above to solve this problem

$$\text{We found that } v = \sqrt{\frac{5}{r}} \tanh(5t * \sqrt{\frac{r}{5}})$$

Instead of limiting the t to 11.5663 due to the drone's limitation, we set the t to infinity

Then, the equation we need to solve for looks like

$$v = \lim_{t \rightarrow \infty} \sqrt{\frac{5}{r}} \tanh(5t * \sqrt{\frac{r}{5}}) = 23.3253004464$$

$$v = \sqrt{\frac{5}{r}}$$

$$v = 23.3253004464$$

We see that this is close to the maximum drone velocity given in the manual, verifying our model

Wind vector and the true direction

Let's say that from (0,0) which is the station and (-300,-400) is the point we need to get to and the wind vector is 10 mph (4.5 m/s) towards Southwest (direction like the Santa Ana winds)

We want to find out how far the drone travels.

Since the time is in seconds,

$$||d|| = 23 \text{ (This is, the drone moves 23 m per second)}$$

$$||w|| = 4.5, \arg(w) = \frac{5\pi}{4} \text{ (wind takes the drone 4.5 m per second, SW direction)}$$

$$||v|| = \sqrt{300^2 + 400^2} = 500$$

Find the angle between the vector w and v

$$\theta = \arccos\left(\frac{w \cdot v}{|w||v|}\right) = 0.14189 \text{ rad} = 8.13 \text{ deg}$$

We now have a SSA triangle, 23m - 4.5m - 8.13°

We find the length of the third side,

$$\frac{\sin(8.13^\circ)}{23} = \frac{\sin(B)}{4.5} \quad B = 1.5856^\circ$$

$$C = 180^\circ - 8.13^\circ - 1.5856^\circ = 170.2844^\circ$$

$$\frac{\sin(8.13^\circ)}{23} = \frac{\sin(170.2844^\circ)}{c} \quad c = 27.446 \text{ m}$$

This means the drone will fly towards the target point at 27.446 m/s

Traveling 500 m at the speed of 27.446 m/s will take **18.2176 seconds** while only using the battery required for 23 m/s

The time it takes and the horizontal distance it travels while ascending and descending

The website says that the ascending speed is 6 m/s and the descending speed is 7 m/s

We assumed already that the wind speed is 4.5 m/s.

If the drone wants to ascend to an altitude of 1000m,

It will do so in $\frac{1000\text{m}}{6 \text{ m/s}} = 166.67$ seconds. It will also travel 750 meters horizontally

If the drone wants to descend from 1000m,

It will do so in $\frac{1000\text{m}}{7 \text{ m/s}} = 142.86$ seconds. It will also travel 642.86 meters horizontally.

Since the time it takes to accelerate to reach 6 m/s is fast, it will be trivial.

Battery calculation with temperature:

Thermal efficiency model

$$T_b = \frac{C \cdot \eta}{P_{total}}$$

T_b = Battery lifetime

C = Battery capacity

η = Battery efficiency factor (varies with temperature),

P_{total} = Total power consumption.

Temperature Effect (η)

$$\eta(T) = 1 - k(T_{nominal} - T)$$

$k = 0.01$ (empirical constant in small temperature deviations)

$T_{\text{nominal}} = 15^{\circ}\text{C}(288\text{K})$.

T = temperature in Kelvin.

Battery performance model:

$$T_b = \frac{C \cdot \eta}{P_{\text{total}}}$$

Because we are only concerned about the battery efficiency factor, we get that

$T_b = T_{\text{ideal}}(1 - k(T_{\text{nominal}} - T))$, given that $\eta(T) = 1 - k(T_{\text{nominal}} - T)$. Assuming that the drone goes up to 1000m altitude, the temperature would be around $8.5^{\circ}\text{C}(281.5\text{K})$. Tidal is 55 minutes for the drone. Plugging in all the above constants, we get that the battery life at the cruising altitude is 51.425 minutes, which is very close to the value of 55 minutes that we were using. Knowing that there are inherent protective systems within batteries to minimize the performance drop, we know that temperature only affects the battery performance in a negligible amount in this scenario.

Results and Observations

Drone Motion Calculations

- The maximum calculated drone velocity was **23m/s**, consistent with manufacturer data
- A drone with a 55-minute battery life has an estimated maximum travel distance of **75.829 km**
- The coverage area per drone was calculated as **18064.27km²**, helping estimate the number of required drone stations for large-scale deployment.

Best Locations for Drone Deployment

- Using weighted geometric median analysis, we identified ideal launch sites for rapid response to fire hazard zones.
- Adjustments for wind effects ensured drones reached target locations efficiently.

Application

The UAVs wildfire detection system

- Professor Zhaodan Kong of the Department of Mechanical and Aerospace Engineering at the University of California, Davis, has developed drone-based methods for wildfire detection and prevention.

- Deploying UAVs equipped with cameras and chemical sensors to monitor abnormal changes in humidity, temperature, and wind, and report potential threats to local fire departments.
- One major challenge in his system is the limited flight duration, which is approximately 30 minutes per drone.
- By applying Professor Kong's detection methodology to our system, which supports **55-minute flights** and covers up to **18,064.27 km²**, we estimate that our drones could monitor **the entire Irvine area** for early wildfire signs.

Post-Wildfire Impact Assessment

- Once deployed, the drone network could also be utilized for **post-wildfire impact analysis**, including assessments of **biological** and **economic damage**.
- Inventors at **Duke University** have developed a drone-based **data relay and mapping system** capable of high-resolution **3D mapping** and **image capture** in remote locations using both small and large UAVs.
- Our **pre-deployed drone stations** can be integrated with Duke University's approach to collect comprehensive post-fire data across affected regions.

Discussion & Conclusion

Research Impact: This modeling presents the optimized framework for deployment in disaster scenarios, improving the response efficiency compared to currently existing methods. Some of the benefits we can obtain by utilizing this model are that the faster the drone detects the fire, the less economic and physical damage there will be, leading to less money cost and loss of innocent lives.

Limitations: Unexpectedly high wind speeds and weather conditions can still affect the drone's performance. In addition, the calculations and results might come out different depending on the size and conditions of regions due to their diverse confounding variables.

Future Work: Future improvements can include more precise calculations on the drone's pathway toward its destination. Another improvement could be in the drone's hardware, enhancing its battery capacity and weight-to-thrust ratio, allowing it to carry more cargo. Future research can also explore new applicable areas for drone networks, particularly in broader contexts such as interdisciplinary fields.